Is a two-scalar hardware signature enough? (Tier 1, item 26)

The two axes are not independent

Across the five machines, in the signature the headline result actually uses:

pair	Pearson r
bandwidth vs **multi-core** compute	**+0.958**
bandwidth vs single-core compute	−0.380
single-core vs multi-core compute	−0.525

Principal components of the standardised (bandwidth, multi-core compute) space:

- axis 1: **97.9%** of the variance
- axis 2: **2.1%** of the variance

The hardware signature is effectively one-dimensional. It is not two scalars in any useful sense — it is one latent machine-quality axis measured twice, at r = 0.958.

What that explains

This is not a curiosity; it accounts for most of what the project has observed.

1. **Why the roofline formula gains so little.** A roofline presumes bandwidth and compute are *independent* resources, so that knowing which one binds tells you something. At r = 0.958 there is no second resource to be bound by, and reweighting between the two terms is interpolating between two estimates of the same quantity. A +1.44pp gain is about what that should buy.
2. **Why the bottleneck classifier is decorative** (`tier1_ablation.md`) and why **measured elasticities do not help as weights either** (`tier1_elasticity.md`). Both findings are forced: there is nothing for a weighting to select between.
3. **Why `1/compute_ratio` alone is the best predictor.** If there is one latent axis, the best predictor is whichever single probe measures it most cleanly — not a blend.
4. **What that axis largely is: core count.** r(multi-core compute, vCPU) = **+0.841**, and r(bandwidth, vCPU) = +0.705. The two 4-vCPU machines (c5a, m5a) sit at the top of both axes. Consistent with the causally confirmed parallelism mechanism (r = 0.458, p = 0.037): PostgreSQL goes parallel, so aggregate capacity is what matters, and aggregate capacity tracks core count.

So, are two scalars enough?

The question cannot be answered on this dataset, and not for want of modelling. Five machines give five points; two axes that are 96% correlated give one effective dimension. There is no room to detect whether a third or fourth axis would help, because the existing two have not been shown to be two.

Settling it requires machines that **decorrelate** the axes — deliberately chosen so high bandwidth does not imply many cores. Concretely: high-core/low-bandwidth and low-core/high-bandwidth instances, arm64 alongside x86, and instance-local NVMe alongside EBS. That is the Tier 2 expansion, and this is the argument for choosing those machines deliberately rather than sampling instance types at random.

The probe suite is built and validated

`scripts/collection/hw_probe_suite.c` measures seven axes instead of two: integer throughput, scalar FP (latency-bound), vector FP, vector FP multi-threaded, unpredictable-branch cost, the L1/L2/L3/DRAM latency hierarchy, and single- and multi-threaded STREAM bandwidth.

It has been run on one machine only — the Apple M1 — because the five EC2 instances are terminated:

probe	M1
int_scalar_gops	3.793
fp_scalar_gflops	1.567
fp_vector_gflops	12.314
fp_vector_gflops_mt	58.467
branch_cost_ns	0.1573
cache_l1_ns	1.293
cache_l2_ns	4.812
cache_l3_ns	7.894
mem_latency_ns	112.051
mem_bandwidth_gbs_1t	55.38
mem_bandwidth_gbs_mt	47.817

Two validation checks pass. `fp_vector_gflops` = 12.314 against `compute_probe.c`'s 12.53 on the same machine (1.8% apart) — two independently written probes agreeing that closely is good evidence both measure what they claim. And the latency hierarchy is strictly monotonic (1.29 → 4.81 → 7.89 → 112.05 ns), which a broken pointer-chase would not produce.

This is infrastructure, not a result. Its value is realised when it runs across the Tier 2 machine set; until then it establishes only that the measurement is ready and sane.

A prerequisite that must be fixed first

The STREAM-triad probe that produced the EC2 bandwidth figures **is not in the repository**. Only `compute_probe.c` survives. So the bandwidth axis — half of the current signature, and the basis of the `naive-linear` baseline every result is compared against — has no reproducible instrument, and no new machine can be placed on it. `hw_probe_suite.c` supersedes it, but on its own scale: numbers it produces are not comparable to the existing 8.44–14.9 figures. Either the whole signature is re-

measured with the new suite, or the old bandwidth numbers stay unreproducible forever. Re-measuring is the only honest option, and it means the five EC2 machines must be relaunched and re-probed as part of Tier 2.